VAYU | YOUR PERSONAL CROP GUIDE

Pomegranate

How to grow it on your land

Prepared for: General guide (no location chosen)
Prepared on 2026-11-09

Suits hot, dry, semi-arid land with drip irrigation, and it earns well. But bacterial blight has made many growers uproot orchards, so disease-free planting material and strict hygiene decide success.

DIFFICULTY	MONEY NEEDED	RISK	FIRST RETURN
4 of 5	**Medium**	**High**	**fruiting from about the 4th year**

How to read this guide: [S1] marks a line backed by a source listed on the last page. (general knowledge) marks a line that has not been verified against a source. Lines starting **For your land** use the numbers for your location.

In this guide

- Your priorities: what to do first, in order
- Your planting plan: how much to buy and prepare for your plot
- How to grow it: planting, water, nutrition, pests and diseases, monitoring, harvest, risks
- If your conditions are different, and the sources behind every line

Your land at a glance

Your priorities

Ordered by what matters most on your land.

1. Talk to your KVK first
Take this guide to your nearest Krishi Vigyan Kendra and ask for their local advice and trusted planting material. (general knowledge)

2. Start small
Plant or build a small trial first. Learn the crop on your own land, then expand from what actually worked. (general knowledge)

3. Line up a buyer before you scale
Know who will buy, in what form and grade, and how it will reach them. (general knowledge)

Your planting plan for 1 acre

What	How much	Basis
Trees at 3 m x 3 m	**about 449**	your plot size x cited amounts [S1]

Sums of the cited per-plant and per-hectare amounts for your plot size. Real needs vary with soil and variety; confirm with your KVK.

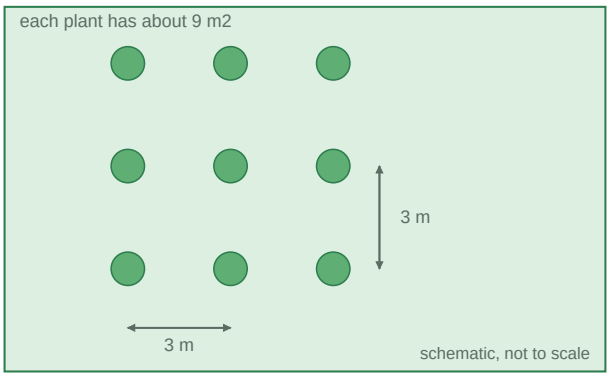
Planting material

- Use certified, disease-free, true-to-type plants free of nematodes, wilt and bacterial blight; tissue-cultured plants from approved sources are one option. ICAR-NRC on Pomegranate (Solapur) is the reference institute. [S1, S8]
- Infected planting material is how bacterial blight reaches new areas, so never buy cheap unverified saplings. [S8]
- Jalore Seedless and G-137 (bred for blight tolerance) are among varieties named, besides Bhagwa; ask your KVK what works locally. [S7, S9]

Land preparation and planting

- Deep alluvial or sandy to sandy-loam soil is best. Plant at about 3 m x 3 m (about 1,111 plants per hectare for Bhagwa); some advice uses wider 4.5 m x 4.5 m to improve airflow. [S1, S6]
- Plant in the monsoon (June-August) or February-March. [S9]

- Mulch the root zone 15-20 cm thick with weeds and dry grass. [S1]

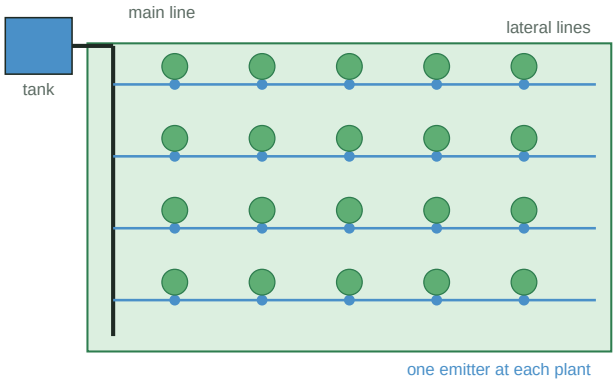


Plant spacing, top view

Irrigation

- Drip irrigation is standard. A drip trial at ICAR-NRCP gave tree water use of about 5.7-13 litres per day in year 2 rising to 10-23 litres in year 5 for Mrig bahar. [S10]

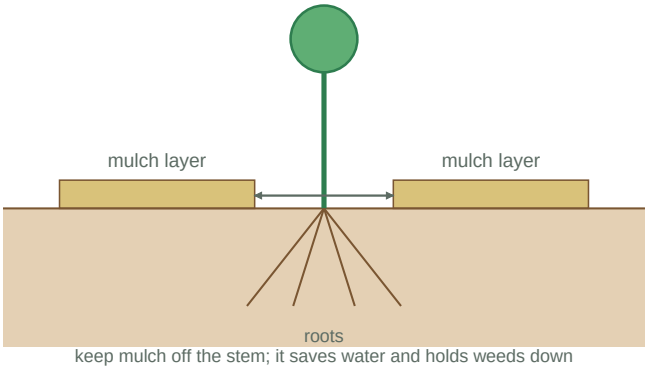
- Crop regulation (bahar) uses planned water withholding, for example April-May for the Mrig bahar crop; learn the schedule from a trained source before trying it. [S11]



Drip irrigation layout (schematic)

Nutrition (organic)

- Farmyard manure or compost around the drip line each year, with mulch; this guide does not give chemical fertiliser doses. Use a soil test and your KVK's schedule. (general knowledge)

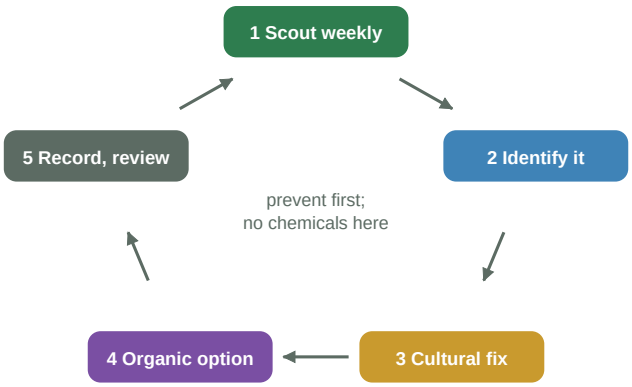


Mulch around a plant

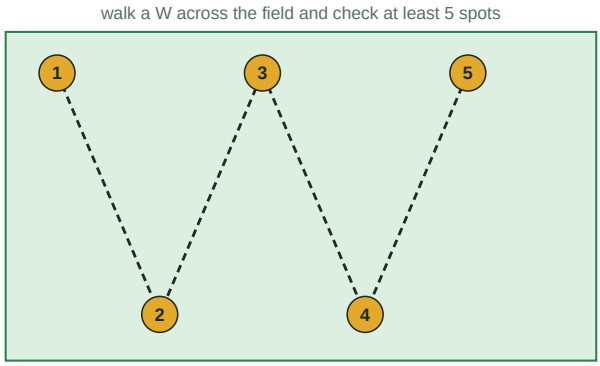
Pests and diseases (organic and cultural only)

- Bacterial blight is the most damaging disease. Prevent with disease-free plants, wide spacing and airflow, and removing infected fruit and twigs and burying them in a covered pit. [S6, S12]
- Biological options studied include Pseudomonas fluorescens and Bacillus subtilis; ask your KVK for a registered biocontrol product and method. [S6]
- Use drip, not sprinklers, so the canopy stays dry; disinfect pruning tools between trees. (general knowledge)
- Institute advisories do list chemical sprays; this guide does not reproduce them. Get the current local schedule from ICAR-NRCP or your KVK. [S12]

- For anything beyond these organic and cultural steps, ask the nearest Krishi Vigyan Kendra (KVK) or state agricultural university for their current recommendation. (general knowledge)



Pest and disease loop: prevent first, then look, fix, review



How to walk the field when scouting

Monitoring

- Walk the orchard weekly: look for dark leaf spots, oily fruit spots, cracking and wilting branches, and remove affected material immediately. (general knowledge)
- Keep a log of flowering, fruit set, irrigation and disease sightings per tree block. (general knowledge)

Harvest, yield and post-harvest

- Fruiting from about the 4th year; harvest timing depends on the bahar (for Mrig bahar, December-January at Solapur). [S13, S11]
- Improved management including blight control raised net returns about 25 percent over farmer practice in an Andhra Pradesh trial of 50 farmers. [S13]

Risks and costs

- A reported 30 percent or more of farmers uprooted trees over four years because of blight, wilt and similar diseases. [S6]
- Water supply and a reliable market and grading are the other big risks; plan both before planting. (general knowledge)

If your conditions are different

What to do if your land, or the season, turns out different from the numbers above.

- **Heat** (fixable): Pomegranate endures up to about 48 C with hot dry winds; fruit colour and quality are best in a hot dry fruiting season. Heat is rarely the limit; water is. [S1]
- **Low rainfall** (fixable): Water is the limiting factor in arid regions; use drip irrigation, mulch 15-20 cm thick, and plan the water budget before planting. [S1, S4, S5]
- **Too much rain** (fixable): Humid, rainy, cloudy fruiting seasons raise disease (bacterial blight, fruit spots) and cracking. Prefer a hot dry fruiting season; choose your flowering season (bahar) to avoid heavy rain on fruit. [S6, S1]

- **Cold weather** (fixable): It is deciduous and tolerates cold winters when irrigated, but young trees can be damaged by hard frost; protect them in the first winters. (general knowledge)
- **Soil pH** (fixable): Ideal pH is about 6.5-7.0, but good orchards in Solapur grow in alkaline soils (pH around 7.8-8.8). Add compost; avoid saline patches. [S1, S7]

About chemicals

This guide gives organic and cultural methods only. It does not name or dose pesticides, fungicides or chemical fertilisers; for those, follow your nearest Krishi Vigyan Kendra (KVK) or state agricultural university and the label of any registered product.

Sources

[S1] https://hpshiva.hp.gov.in/cms/media/ggojam4f/pomegranate.pdf
[S4] https://epubs.icar.org.in/index.php/IJAgS/article/download/65400/27693/164294
[S5] https://www.phytojournal.com/archives/2020/vol9issue1/PartH/8-3-661-262.pdf
[S6] https://www.masujournal.org/110/Article_21_1097_cHANDANA.pdf
[S7] https://journal.iahs.org.in/index.php/ijh/article/download/164/1450
[S8] https://epubs.icar.org.in/index.php/IJAgS/article/download/102150/40011/340517
[S9] https://indianagriexam.com/?p=1810
[S10] https://journal.iahs.org.in/index.php/ijh/article/download/1599/1092
[S11] https://epubs.icar.org.in/index.php/IJAgS/article/download/102150/40011/340517
[S12] https://nrcpomegranate.icar.gov.in/files/Advisory/71.pdf
[S13] https://www.phytojournal.com/archives/2020/vol9issue1/PartH/8-3-661-262.pdf

Advisory aid from published sources and general agronomic knowledge, not a guarantee of yield or income and not a substitute for local extension advice. Lines marked 'general knowledge' are not verified against a source.